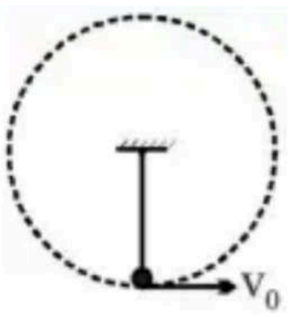
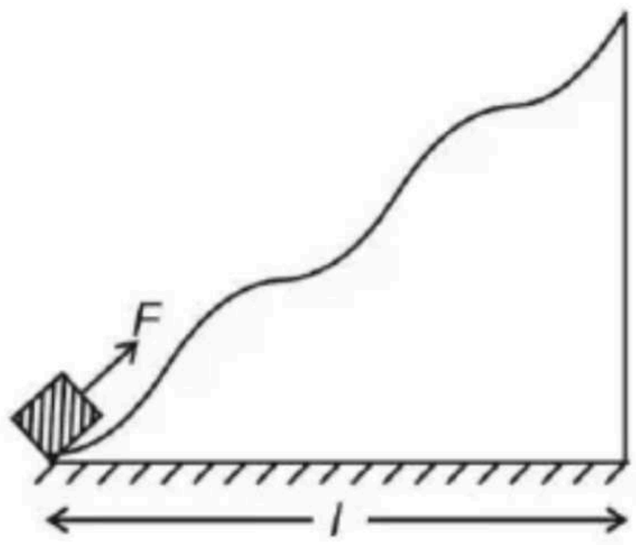
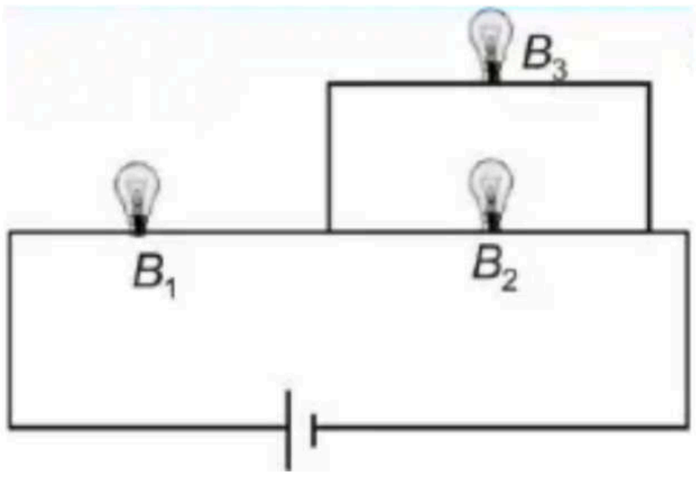
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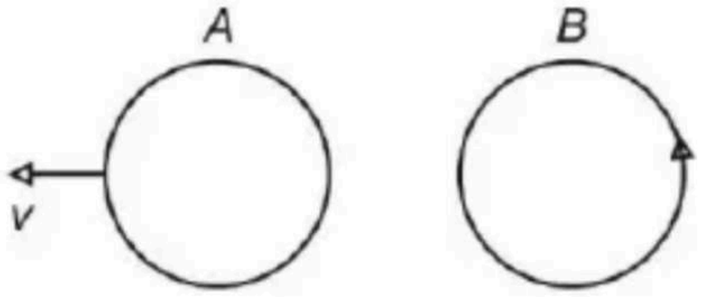
THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

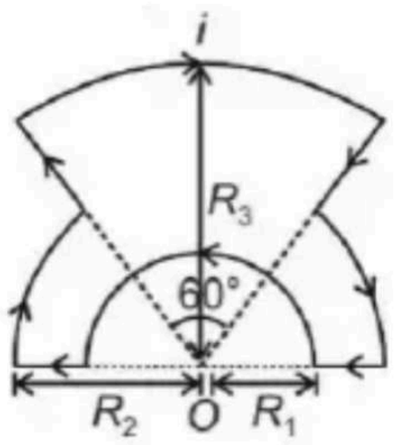
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V_0


F
l


B_3
B_1
B_2


A
B
v


i
R_3
60°
R_2
O
R_1

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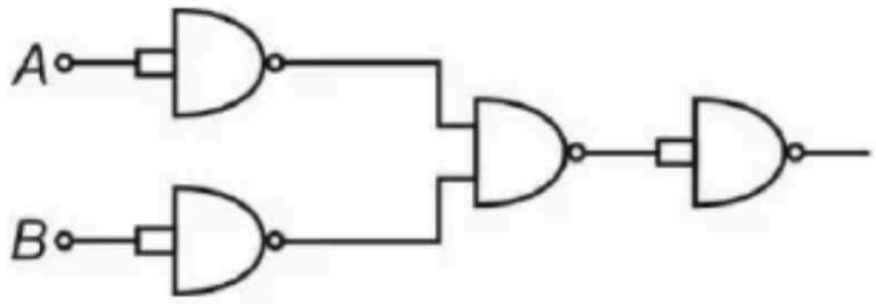
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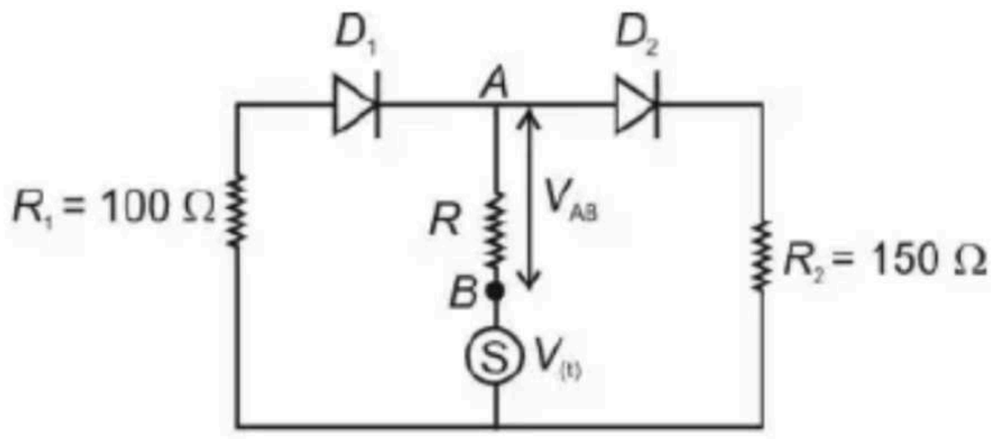
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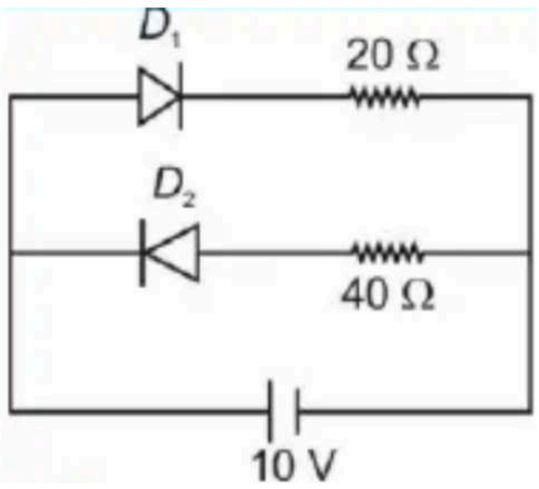

A
B

—

—


D1
D2
A
R1 = 100 Ω
R
VAB
B
S
V(t)
R2 = 150 Ω

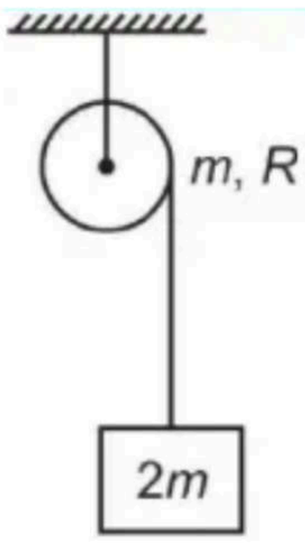
35. Two ideal diodes are connected to a battery as shown in the circuit. The current supplied by the battery is



(1) Zero (2) 0.5 A

(3) 0.4 A (4) 0.75 A

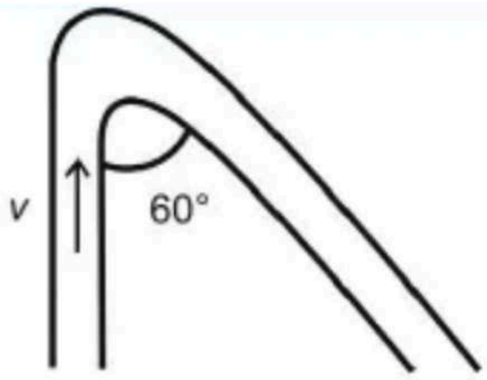
36. A block of mass 2m is connected with a string which is wound over a pulley of mass m and radius R as shown in figure. If the system is released from rest, then angular acceleration of the pulley will be



(1) — (2) —

(3) — (4) —

37. Water (density) is flowing through the uniform tube of cross-sectional area A with a constant speed as shown. The magnitude of force exerted by the water on the curved portion of the tube is



(1) √ (2)

(3) √ (4) √

38. Two soap bubble coalesce to form one bubble. If decreases in volume and surface area are V and S respectively. Atmospheric pressure is P_0 and surface tension is T, then correct relation is :

(1) (2)

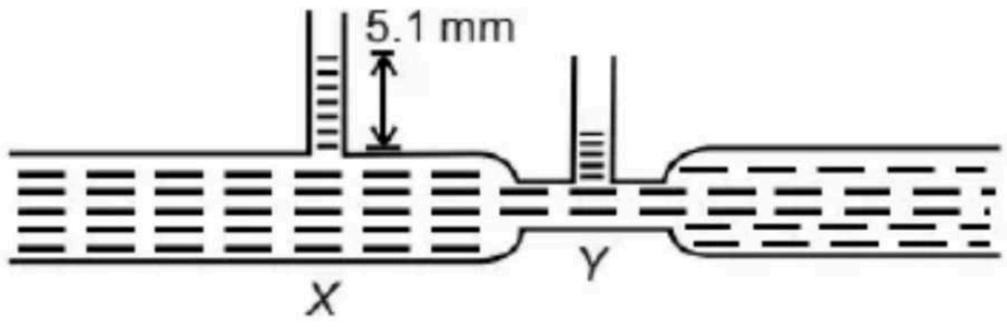
(3) (4) None of these

39. Find the heat current flowing through a hollow thick sphere of inner radius 'R' and outer radius '2R'. Thermal conductivity = K. Temperature inside is and outside is .

(1) (2)

(3) (4) ( )

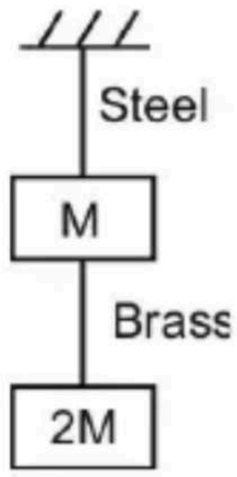
40. A diagram shows a venturimeter through which water is flowing The speed of water at X is The speed of water at Y is ($g = 1000\ s^{-2}$)



(1 (2

(3 (4

41. If the ratio of lengths, radii and Youngs moduli of steel and brass wires in the figure are a, b and c respectively, then the corresponding ratio of increase in their lengths is



(1) — (2) —

(3) — (4) —

42. A graph is plotted by taking the temperature in Fahrenheit along the y-axis and the corresponding temperature in kelvin along the x-axis.

Which among the following statements (is/are) correct?

(A) Shape of curve is parabolic

(B) Shape of curve is straight line

(C) Slope of curve at a point may be $\frac{19}{5}$

(D) At 0 K temperature reading of Fahrenheit scale is -459.7° F.

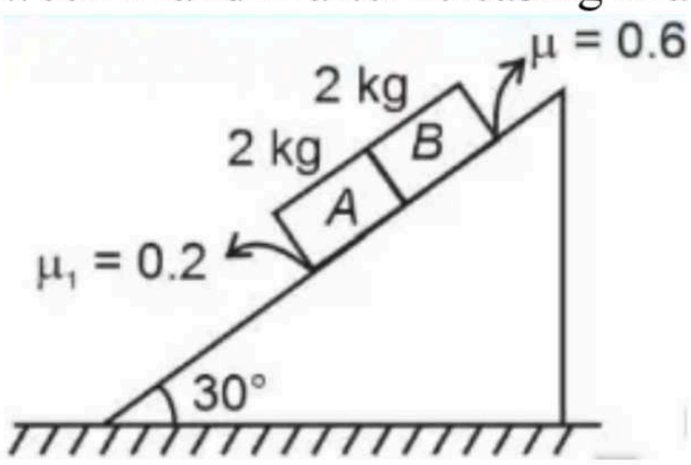
(1) (B) and (C) only

(2) (B) and (D) only

(3) (A) and (D) only

(4) (A), (C) and (D) only

43. In the diagram given below, normal reaction between A and B after releasing A and B is



(1) Zero (2) 10 N

(3) 20 N (4) 30 N

44. In the following question, two statements A and B are given.

Choose correct option according to given statements.

Statement A: Newton was inspired to arrive at universal law of gravitation that led to an explanation of Kepler's laws.

Statement B: Newton's reasoning for moon revolving around earth was that earth provides gravitational pull and is subjected to centripetal acceleration.

(1) Statement A is correct but statement B is incorrect

(2) Both the statements A and B are correct

(3) Statement A is incorrect but statement B is correct

(4) Both the statements A and B are incorrect

45. Acceleration experienced by a mass m at height h = $2R_E$ above earth surface due to gravitational pull of earth is (Take M as mass of earth and R_E its radius)

(1) $\frac{GM}{3R_E^2}$ (2) $\frac{GM}{6R_E^2}$

(3) $\frac{GM}{9R_E^2}$ (4) $\frac{GM}{4R_E^2}$

CHEMISTRY

46. Given below are two statements.

Statement – I : The freezing point of 0.1 M aqueous solution of urea is less than that of 0.1 M aqueous solution of NaC*l*.

Statement – II : NaC*l* is an electrolyte and aqueous solution Contains more number of particles.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statements, I and II are incorrect.

(2) Both statements, I and II are correct

(3) Statement I is incorrect, but II is correct

(4) Statement I is correct, but II is incorrect

47. Liquid P and Q form an ideal solution at 300 K, the vapour pressure of pure P and Q are 200 Pa and 500 Pa respectively, The composition of P in the Vapours which is in equilibrium with a solution Containing 50 mole percent of P at the same temperature is

(1) 28.5 % (2) 71.5 %

(3) 35.6 % (4) 47.75 %

48. The limiting molar conductance (λ^0) of KBr, KCl and NaCl are 152,150 and 126 respectively. Then, λ^0 of NaBr will be-

(1) 102 (2) 128

(3) 167 (4) 12

49. What will be the emf (E) of given cell?

$$Pt, \underset{P_1}{H_{2(g)}} \left| \underset{C_1}{H^+_{(eq)}} \right|\left| \underset{C_2}{H^+_{(eq)}} \right| \underset{P_2}{H_{2(g)}} \left| Pt \right.$$

(1) — — (2) $\frac{RT}{2F}\ln\frac{C_2^2P_1}{C_1^2P_2}$.

(3) — — (4) — —

50. Match the Column. I with Column II. and choose the correct option

Column I		Column II	
(P)	1 mole of $Cr_2O_7^{-2}$ to Cr^{+3}	(I)	193000 C
(Q)	2 mole of Pb^{+4} to Pb^{+2}	(II)	579000 C
(R)	1 mole of MnO_4^- to Mn^{+2}	(III)	386000 C
(S)	1 mole of $C_2O_4^{-2}$ to CO_2	(IV)	482500 C

Choose the most appropriate from the options given below.

(1) P – II, Q–III, R–IV, S–I

(2) P – II, Q–I, R–IV, S–I

(3) P – IV, Q–III, R–II, S–I

(4) P – I, Q–II, R–III, S–IV

51. In a gas phase reaction $X_{(g)} \longrightarrow Y_{(g)} + 2Z_{(g)}$, the increase in pressure from 100 mm of Hg to 120 mm of Hg is noticed in 5 minutes. If decomposition of **X** follows first order kinetics then what is the rate of decomposition of X at 10th min ?

(1) $1.695 mm\,(Hg)\,min^{-1}$

(2) $2.318\,mm(Hg)min^{-1}$

(3) $3.679\,mm(Hg)min^{-1}$

(4) $2.875\,mm(Hg)min^{-1}$

52. Given below are two statements.

Statement – I : With increase in temperature rate of reaction increases

Statement – II : Increasing the temperature of reaction increases the fraction of molecules Which collide with energy greater than Threshold Energy.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statements, I and II are incorrect.

(2) Both statements, I and II are correct

(3) Statement I is incorrect, but II is correct

(4) Statement I is correct, but II is incorrect

53. Choose the correct statements.

(1) NH_3 is less basic than PH_3 because N is more EN than P.

(2) The correct order of B.P. of hydride is $NH_3 > SbH_3 > AsH_3 > PH_3$

(3) PH_3 is most volatile hydride in group VA.

(4) NO is colorless, diamagnetic gas

54. How many statements given below are Correct ?

(1) Chorine has highest electron-affinity.

(2) Fluorine oxidises water to oxygen

(3) Concentrated H_2SO_4 Can not be used to prepare HBr from KBr.

(4) XeF_6 reacts with RbF to form $Rb^+[XeF_7]^-$

(5) SO_3 contains only one $P_\pi - d_\pi$ bond.

(1) 1,2,3 only (2) 1,3,4,5 only

(3) 1,2,3,4 only (4) 1,2,3,4,5

55. Which of the following transition element does not show variable oxidation state?

(1) Cr (2) Mn

(3) Ti (4) Sc

56. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : $KMnO_4$ is purple in colour.

Reason (R) : The purple colour of $KMnO_4$ is due to d-d electron transition

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false

57. Which of the following has lowest C– O bond order?

(1) $[Cr(CO)_6]$ (2) $[Mn(CO)_6]^+$

(3) $[V(CO)_6]^-$ (4) $[Fe(CO)_5]$

58. Choose the correct statement among the following

(A) The oxidation number of Ni in $[Ni(DMG)_2]$ is Zero.

(B) Na_2 EDTA is used to remove Ca^{+2} and Mg^{+2} ions from hard water.

(C) Vitamin B-12 contains Co^{+3}

(D) Cis-platin is used for treatment of cancer

(1) A, B, D only (2) A, D only

(3) A, C, D only (4) A, B, C, D.

59. The set of II group basic radicals whose yellow ppt are soluble in yellow ammonium sulphide is -

(1) $Pb^{+2}, Bi^{+3}, As^{+3}$ (2) $Hg^{+2}, Pb^{+2}, Sn^{+4}$

(3) $As^{+3}, Sn^{+4}, Cd^{+2}$ (4) $Cu^{+2}, Sn^{+2}, As^{+3}$

60. Given below are two statements.

Statement-I: Acetaldehyde responds positively with Fehling test, Tollen's test, 2,4- DNP test, as well as iodoform test.

Statement-II: All aldehyde responds all 4 tests mentioned in statement-I.

In light of the above statements, choose the correct answer from the options given below.

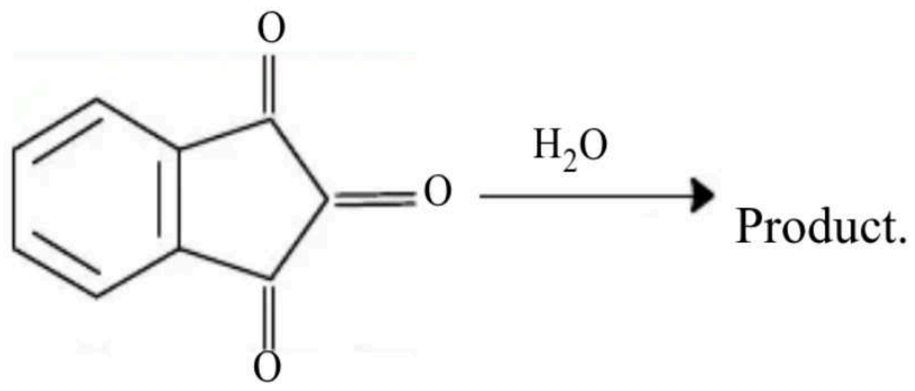
(1) Both statements, I and II are incorrect.

(2) Both statements, I and II are correct

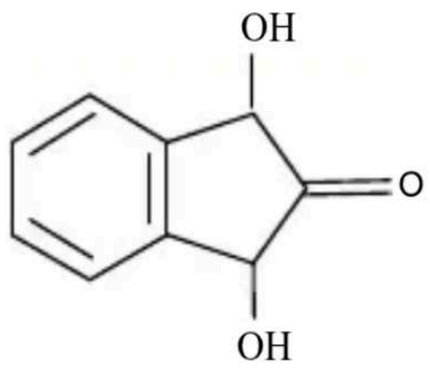
(3) Statement I is incorrect, but II is correct

(4) Statement I is correct, but II is incorrect

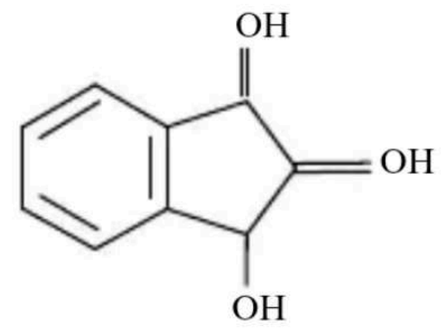
61. In the following reaction, the major product is



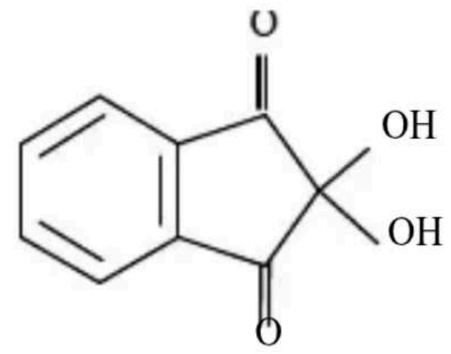
(1)



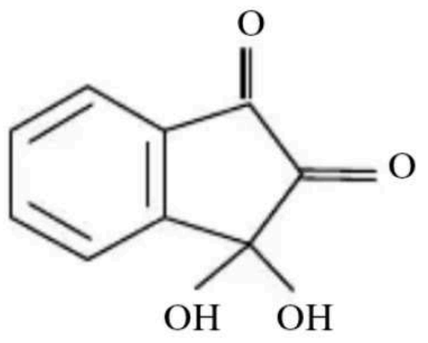
(2)



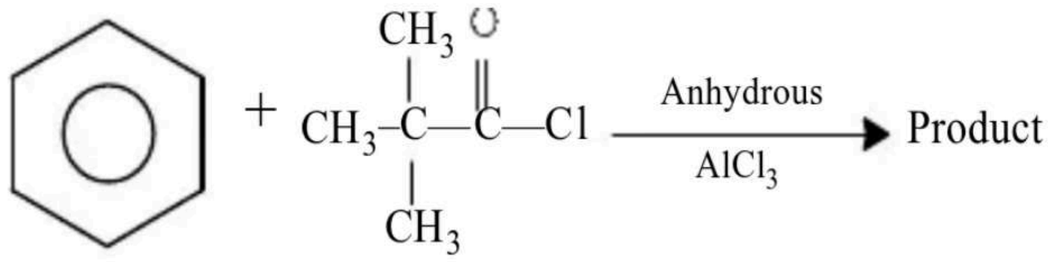
(3)



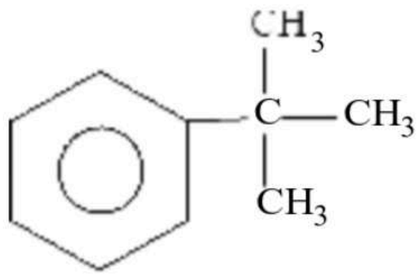
(4)



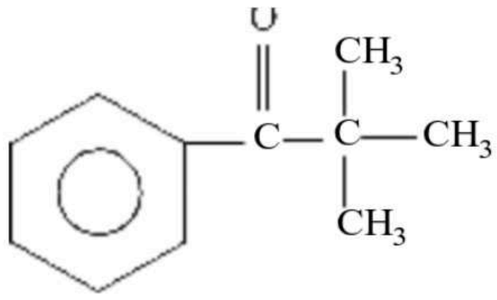
62. What is the Major product in the following reaction?



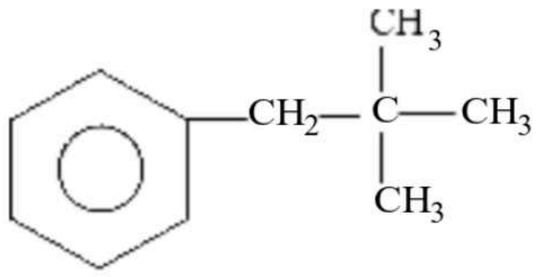
(1)



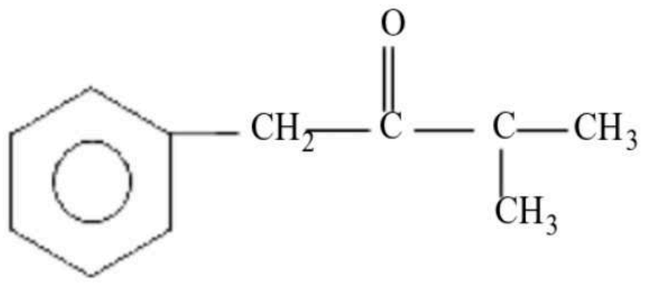
(2)



(3)



(4)



63. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : Phenol is more acidic than benzoic aid because phenoxide ion is more stable than benzoate ion.

Reason (R) : In phenoxide ion, negative charge is delocalised over one oxygen and three carbon atoms so it is more stable than benzoate ion

In light of the above statements, choose the correct answer from the options given below.

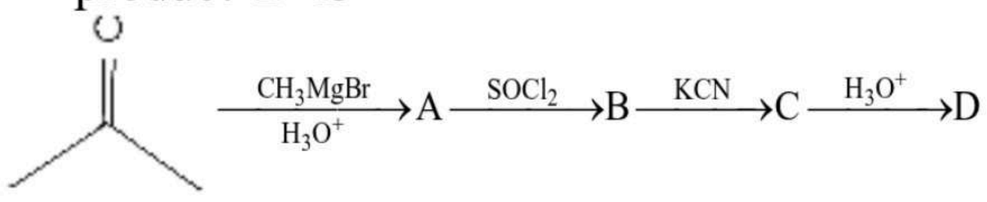
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) Both (A) and (R) are incorrect.

64. In the given reaction sequence, the final product 'D' is –



(1) 2,2,-dimethyl butanoic acid.

(2) 3-Ethyl-2-methylpropanoic acid.

(3) 2-methyl butanoic acid.

(4) 2,2-dimethylpropanoic acid.

65. Choose the correct statement about the following sequence of reaction.

$CH_3 - CHCl_2 \xrightarrow{aq\cdot NaOH} A \xrightarrow{I_2+NaOH} B + CHI_3$

(A) B is sodium formate.

(B) B is treated with dilute HCl followed by $LiAlH_4$ & H_2O gives methanol.

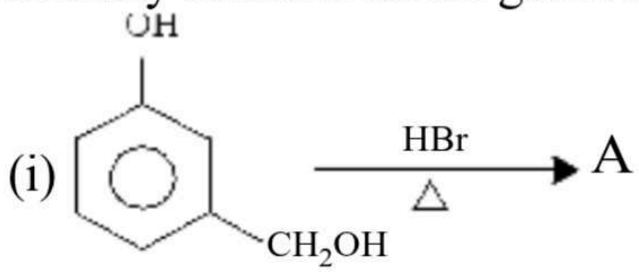
(C) A is acetaldehyde.

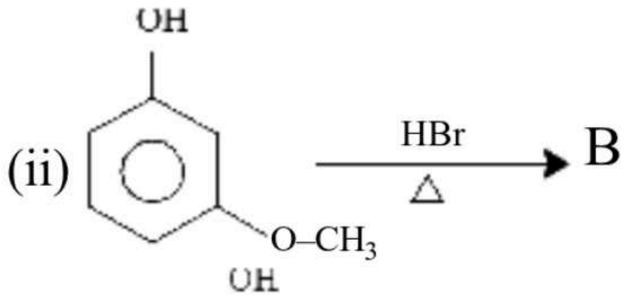
(D) On treatment with NH_2OH, A gives two products.

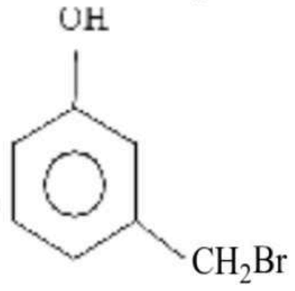
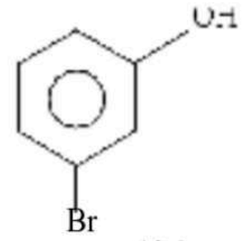
(1) A, B, only (2) A & C only

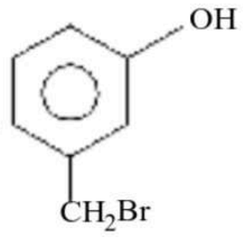
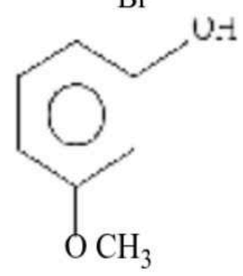
(3) A, B, C only (4) A, B, C, D

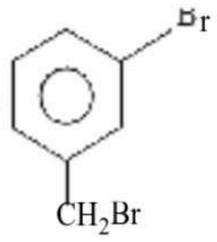
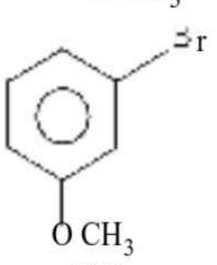
66. Identify A and B in the given reaction.

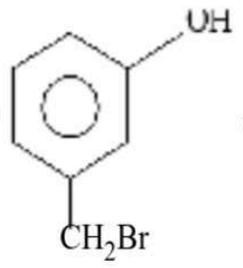
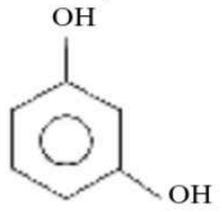




(1) A is  , B is 

(2) A is  , B is 

(3) A is  , B is 

(4) A is  , B is 

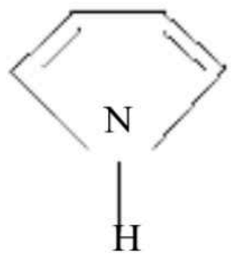
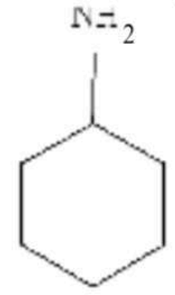
67. The correct order of reactivity of following compounds with $CH_3 - O^{\ddot{-}}$ in an SN^2 reaction is

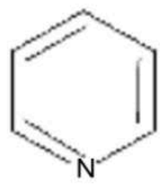
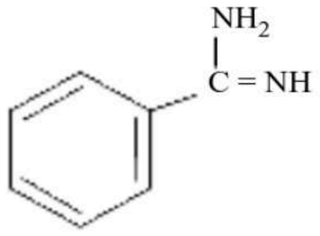
(1) $CH - Br > CH_3 - F > CH_3 - OH > CH_3OSO_2CF_3$

(2) $CH_3 - O.SO_2CF_3 > CH_3 - Br > CH_3 - OH > CH_3 - F$

(3) $CH_3 - OSO_2CF_3 > CH_3 - Br > CH_3 - F > CH_3 - OH$

(4) $CH_3OH > CH_3 - Br > CH_3 - F > CH_3OSO_2CF_3$

68. (I)  (II) 

(III)  (IV) 

(P) I and III are aromatic and III is more basic than I

(Q) I is antiaromatic, III is aromatic and III is more basic than I

(R) Basic strength order is I < III < II < IV

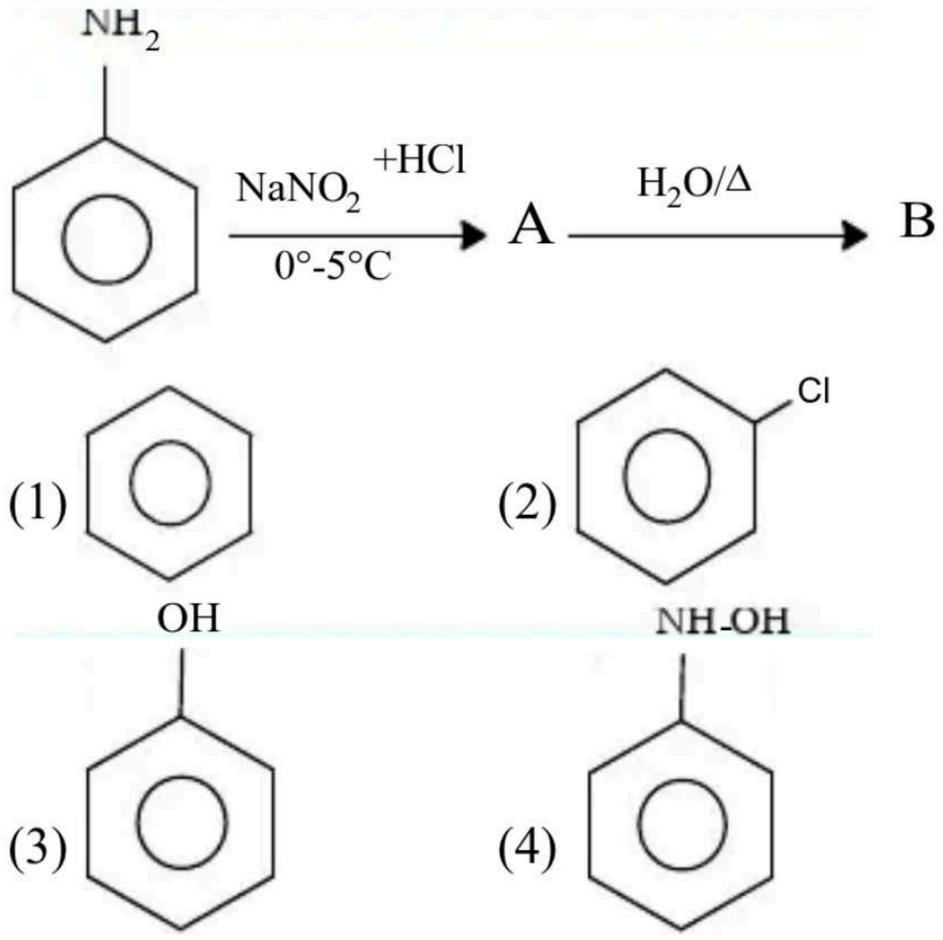
(S) Conjugated acid of III is less stabilised than Conjugated acid of IV

Choose the correct statement out of given

(1) P, Q, R only (2) P, R only

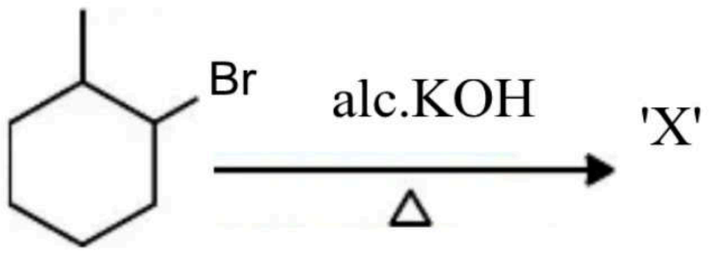
(3) R & S only (4) P, R, S only

69. What is B?



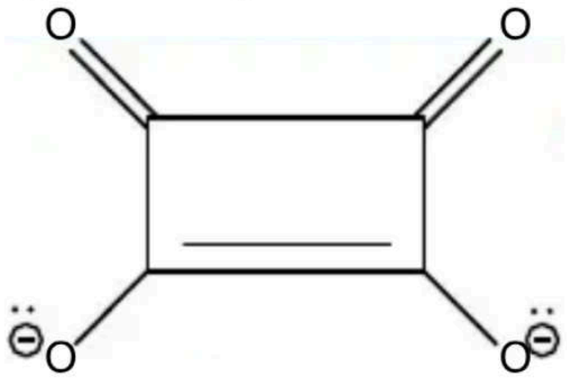
70. Given below are two statements.
Statement-I: All monosaccharides are reducing sugar.
Statement-II: D(+) glucose and L(–) glucose are enantiomers
In light of the above statements, choose the correct answer from the options given below.
(1) Both statements, I and II are incorrect.
(2) Both statements, I and II are correct
(3) Statement I is incorrect, but II is correct
(4) Statement I is correct, but II is incorrect

71. The number of C–H bond that Can involve in hyperconjugation in the major product x in the following reaction is –



(1) 3 (2) 4
(3) 5 (4) 7

72. How many resonating structures are possible for the given compound?



(1) 2 (2) 4
(3) 5 (4) 10

73. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).
Assertion (A) : Cis-1-chloropropene has higher boiling point than Trans-1-chloropropene.
Reason (R) : Cis-1-chloropropene has higher dipole moment compared to Trans-1-chloropropene.
In light of the above statements, choose the correct answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) Both (A) and (R) are false

74. The hybridisation of Al in the $\left[Al(H_2O)_6\right]^{+3}$ ion is
(1) sp^3 (2) dsp^2
(3) sp^3d (4) sp^3d^2

75. $AlCl_3$ is an electron deficient compound but AlF_3 is not because
(1) Atomic size of F is smaller than Cl which makes AlF_3 more covalent.
(2) Al in AlF_3 is sp^2 hybridised while in $AlCl_3$ it is sp^3 hybridized
(3) $AlCl_3$ exists as dimer but AlF_3 does not
(4) $AlCl_3$ is covalent compound while AlF_3 is an ionic compound.

76. Choose the correct increasing order of energy of the orbitals having following set of quantum number
P– n = 2, ℓ = 0, Q–n = 3, ℓ = 0
R–n = 2, ℓ = 1 S –n = 3, ℓ = 1
(1) P < Q < R < S (2) P < R < Q < S
(3) S < Q < R < P (4) P < R < S < Q

77. Choose the Correct Code. Containing correct statements out of given.

(A) A spectral line will not be seen for a $2P_x - 2P_y$ transition.

(B) As the intensity of incident light increases, photo current increases

(C) Velocity of electron increases with decrease in principal quantum number.

(D) Energies of the orbitals in the same subshell decrease with increase in atomic Number

(1) A & B only (2) A & C only

(3) A, B & D only (4) A, B, C & D

78. Match the column I & II.

	Column I		Column II
(A)	Most reactive metal	(I)	ns^2np^5
(B)	Inner transition metal	(II)	ns^2np^1
(C)	Most electron affinitive element	(III)	$(n-2)f^7$ $(n-1)d^1ns^2$
(D)	Amphoteric metal	(IV)	ns^{1-2}

Choose the most appropriate from the options given below.

(1) A –I, B–II, C-III, D-IV

(2) A –IV, B–III, C-I, D-II

(3) A –IV, B–III, C-IV, D-I

(4) A –II, B–III, C-IV, D-I

79. The correct increasing order of Basic strength is

(1) $MgO < Cs_2O < Na_2O < Al_2O_3$

(2) $Al_2O_3 < Na_2O < MgO < CS_2O$

(3) $Al_2O_3 < MgO < Na_2O < Cs_2O$

(4) $Cs_2O < Na_2O < MgO < Al_2O_3$

80. How many of the following have planar geometry?

$CCl_4, SO_3, SF_4, XeF_4, BF_3, CH_2 = CH_2, PCl_5$

(1) 2 (2) 3

(3) 4 (4) 5

81. In which of the following, the bond order has increased, and the magnetic behavior has changed.?

(1) $C_2^+ \longrightarrow C_2$ (2) $NO^- \longrightarrow NO^+$

(3) $N_2^+ \longrightarrow N_2$ (4) all of these.

82. If Enthalpy of neutralisation of CH_3COOH with NaOH is $-11.3 Kcal/mole$, then the enthalpy of ionisation of CH_3COOH is -

(1) – 1.3 Kcal/ mole

(2) –2.31.3 Kcal/ mole

(3) +1.3 Kcal/mole

(4) +2.3 Kcal/mole

83. Consider the statements given and find. How many are incorrect?

(A) Solubility of sulphates of alkaline earth metals increases down the group.

(B) BeF_2 is highly soluble in water

(C) Increasing order of bond angle is $H_2O < F_2O < Cl_2O$

(D) Bond order of CO is 3 and of CO^+ is 2.5

(E) Xe in XeF_2 is dsp^2 hybridised.

(1) A, B, C, D only (2) A, B, D, E only

(3) A, C, E only (4) A, C, D, E only

84. The correct statement is -

(A) Salicylic acid has intramolecular H-bonding

(B) NH_3 has high enthalpy of vaporisation due to intermolecular H–bonding

(C) $CuSO_4.5H_2O$ has ionic, covalent, Coordinate and hydrogen bonds.

(D) Be is Amphoteric metal and does not impart colour to the flame.

(E) The s-character in Diamond, graphite and acetylene is 50%, 33.3% & 25% respectively

(1) B & C only (2) A, B, E only

(3) B, C, D, E only (4) A, B, C, D only

85. The heat liberated on Complete combustion of 7.8 g of benzene is 327 KJ . This heat was measured at constant volume and at 27°C Calculate the heat of combustion of benzene at constant pressure

(1) $-3274\,kJ/mole$ (2) $-1637\,kJ/mole$

(3) $-3270\,kJ/mole$ (4) $-3637\,kJ/mole$

86. The pH of a buffer solution made by mixing 50 ml of $0.01M\,NH_4OH$ and 50 ml of $0.10M\,NH_4Cl$ is $(PK_b$ of $NH_4OH = 4.8)$

(1) 4.8 (2) **6.2**

(3) 5.8 (4) 8.2

87. The partial pressure of CO and CO_2 in the given reaction at equilibrium is 8 and 4 atm. respectively. What is the value of K_p for this reaction ?

$C_{(s)} + CO_2(g) \rightleftharpoons 2CO_{(g)}$

(1) 8 (2) 16

(3) 32 (4) 64

88. 200 Kg of 60% pure $CaCO_3$ is heated strongly to decompose into CaO and CO_2.

Find the volume of CO_2 gas at STP obtained.

(1) $2.68\times10^4\,L$ (2) $4.48\times10^4\,L$

(3) $2.24\times10^4\,L$ (4) $1.12\times10^4\,L$

89. Choose the correct statement –

(A) Water acts as base with H_2S and as acid with NH_3

(B) Equilibrium constant for dissociation of water (K_c) is 1.8×10^{-16}

(C) pH of 0.01 M aqueous solution of NaOH at 25° is 12

(D) Solution having $[H]^+$ equal to 10^{-9} m is basic in nature

(E) The $[OH]^-$ in 0.05 M H_2SO_4 is 1×10^{-12}

(1) B, C, E only (2) A,B,C,E only

(3) A, C, D,E only (4) A,B,C,D,E only

90. Concentration of the Ag^+ ions in a saturated solution of $Ag_2C_2O_4$ is $2.2\times10^{-4}\,molL^{-1}$. Solubility product of $Ag_2C_2O_4$ is :-

(1) 2.66×10^{-12} (2) 4.5×10^{-11}

(3) 5.3×10^{-12} (4) 2.42×10^{-8}

BIOLOGY

91. The order Primata and Carnivora belong to:

(1) the same family called Mammalia.

(2) the same class called Mammalia.

(3) the same division called plantae.

(4) the same kingdom called angiosperm

92. The two polypeptide A and B chains of human immature insulin are linked together by

(1) hydrogen bonds

(2) phosphodiester bond

(3) covalent bond

(4) disulphide bridges

93. Which of the following features is *not* represented in the phylum Arthropoda?

(1) Chitinous exoskeleton

(2) Metameric segmentation

(3) Parapodia

(4) Jointed appendages

94. The chronic respiratory disorder caused by cigarette smoking is

(1) emphysema (2) asthma

(3) silicosis (4) pneumoconiosis

95. Which one of the following is common among,

Cyanobacteria, Archaebacteria, Eubacteria

(1) Presence of introns

(2) Heterotrophic nutrition

(3) Lipo-proteinaceous cell membrane

(4) More than one correct

96. Asthma may be attributed to

(1) bacterial infection of the lungs

(2) allergic reaction of erythrocytes cells in the lungs

(3) inflammation of bronchi and bronchioles

(4) accumulation of fluid in the lungs

97. Reserve food in phaeophyceae and phycomycetes is respectively:
(1) Starch and glycogen
(2) Glycogen and laminarin
(3) Mannitol and glycogen
(4) Laminarin and mannitol

98. Which of the following plants has the following characters.

Some thrive in sandy soil condition, non-vascular dioecious gametophyte, presence of small sized leaves

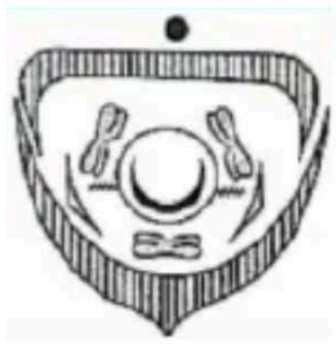
(1) *Adiantum* (2) *Dryopteris*
(3) *Selaginella* (4) *Pteridium*

99. Select the *incorrect* statement.
(1) FSH stimulates the Sertoli cells which help in spermiogenesis.
(2) LH triggers ovulation in the ovary.
(3) LH and FSH decrease gradually during the follicular phase.
(4) LH triggers secretion of androgens from the Leydig cells.

100. In higher vertebrates, the immune system can distinguish self and non-self. If this property is lost due to genetic abnormality and it attacks self-cells, then it leads to
(1) allergic response
(2) graft rejection
(3) autoimmune disease
(4) active immunity

101. Which of the following is *incorrect* regarding vasectomy?
(1) Vasa deferentia is cut and tied
(2) Terminal method of contraception
(3) No sperm occurs in seminal fluid
(4) No sperm occurs in epididymis

102. Which plant family and its pair of examples is correctly relates with this floral diagram:



(1) Liliaceae - *Colchicum* and Garlic
(2) Asteraceae – Sunflower, *Tagetes*
(3) Malvaceae – Cotton, Lady finger
(4) Poaceae - Wheat and Sugarcane

103. In the racemose type of inflorescence:
(a) The main axis terminates in a flower.
(b) The flowers are born in acropetal succession.
(c) The younger flowers are born at the base of the floral axis.
Select the ***incorrect*** statements.
(1) Only (a) (2) (a) and (b)
(3) (a) and (c) (4) (a), (b) and (c)

104. Assertion(A): Rhizome is a stem modification.
Reason(R): It grows horizontally underground and bears nodes and internodes.
(1) Both (A) and (R) are true, and (R) is correct explanation of A.
(2) Both (A) and (R) are true, but (R) is not the correct explanation of A.
(3) (A) is true, but (R) is false.
(4) Both (A) and (R) are false statements.

105. Statement I : Collenchyma provides mechanical support to the growing parts of the plant.
Statement II: Its cells possess unevenly thickened cell walls ,much thickened at the corners
(1) Both statements are correct.
(2) Both statements are incorrect.
(3) Statements I is correct, and II is incorrect.
(4) Statements I is incorrect, and II is correct.

106. Match the following and select the correct answer:

List – I	List – II
A. Centriole	(i) Infoldings in mitochondria
B. Chlorophyll	(ii) Thylakoids
C. Cristae	(iii) Nucleic acids
D. Ribozymes	(iv) Basal body of cilia or flagella

(1) A : iv, B : ii, C : i, D : iii
(2) A : i, B : iii, C : ii D : iv
(3) A : i, B : ii, C : iii, D : iv
(4) A : iv, B : iii, C : i, D : ii

107. Which of the following approaches does *not* give the defined action of contraceptive methods?

(1)	Barrier methods	Prevent fertilisation
(2)	Intrauterine devices	Increase phagocytosis of sperm, suppress sperm motility and fertilising capacity of sperms
(3)	Hormonal contraceptives	Prevent ovulation and implantation
(4)	Vasectomy	Prevents spermatogenesis

108. Antivenom injection contains preformed antibodies while polio drops that are administered into the body contain

(1) activated pathogens

(2) harvested antibodies

(3) gamma globulin

(4) attenuated pathogens

109. In Hardy-Weinberg equation, the frequency of heterozygous individuals is represented by

(1) pq (2) q^2

(3) p^2 (4) 2pq

110. How many among the following are correctly matched?

(a) Pili - Play a role in motility

(b) Fimbriae - Helps bacteria to attach on rocks and to the host tissue

(c) Chromatophore -Present in cyanobacteria

(d) Glycocalyx - Composition differs among different bacteria

(1) One (2) Two

(3) Three (4) Four

111. Which of the following is correct regarding AIDS causative agent HIV?

(1) HIV is an unenveloped retrovirus.

(2) HIV spreads by mere touch or physical contact.

(3) HIV is an enveloped virus containing one molecule of single-stranded RNA and one molecule of reverse transcriptase enzyme.

(4) HIV is an enveloped virus that contains two identical molecules of single-stranded RNA and two molecules of reverse transcriptase enzyme.

112. Which of the following characteristic features always holds true for the corresponding group of animals?

(1)	Cartilaginous endoskeleton	Chondrichthyes
(2)	Viviparous	Mammalia
(3)	Possess a mouth with an upper and a lower jaw	Chordata
(4)	3-chambered heart with one incompletely divided ventricle	Reptilia

113. Select correct option w.r.t cell:

(1) Mitochondria is the largest cytoplasmic organelle of plants.

(2) Ribosome is smallest non membranous cell organelle in plants.

(3) The chloroplast is double membrane bound organelle and the inner chloroplast membrane is relatively more permeable.

(4) Thylakoids are arranged in stacks like the piles of coins called stromal thylakoids/ stroma lamellae.

114. Choose the correct statement.

(1) All reptiles have a three-chambered heart.

(2) All Pisces has gills covered by an operculum.

(3) All mammals are viviparous.

(4) All cyclostomes do not possess jaws and paired fins.

115. In cockroach, alary muscles are associated with

(1) heart and circulation

(2) Malpighian tubule and excretion

(3) trachea and respiration

(4) hepatic caecae and digestion

116. In spermatogenesis, four daughter cells are produced by two successive meiotic divisions in which
(1) first division is equational and second is reductional
(2) first division is reductional and second is equational
(3) both divisions are equational
(4) both divisions are reductional

117. Assertion: Process of the meiosis is, per se, paradoxical.
Reason: Meiosis increases the genetic variability in the population of organisms from one generation to the next.
(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) Both (A) and (R) are false.

118. Red skeletal muscle fibres are red because of the presence of
(1) myoglobin and mitochondria
(2) haemoglobin and golgi bodies
(3) globulin and mitochondria
(4) protein and lysosome

119. DNA replication occurs during _A_ and increase in number of chromosomes is observed first during _B_. Identify A and B from following options
(1) A-Interphase, B-Prophase
(2) A-S-phase, B-Telophase
(3) A-Synthetic phase, B-Gap 2 phase
(4) A-Interphase, B- Anaphase

120. The concept of chemical evolution is based upon
(1) possible origin of life by combination of chemicals under suitable environmental conditions
(2) crystallisation of chemicals
(3) interaction of water, air and clay under intense heat
(4) effect of solar radiation on chemicals

121. Match the Column I with Column II and choose the correct option.

Column I	Column II
A) Hyaluronidase	1. Acrosomal reaction
B) Corpus luteum	2. Morphogenetic movements
C) Gastrulation	3. Progesterone
D) Capacitation	4. Milk produced during lactation
E) Colostrum	5. Sperm activation

(1) A-5, B-2, C-4, D-1, E-3
(2) A-1, B-3, C-2, D-5, E-4
(3) A-2, B-3, C-4, D-5, E-1
(4) A-1, B-2, C-3, D-5, E-4

122. If a tetraploid (4n) plant cell in the G_1 phase has a DNA content of 4C, match the description/stage in list-I with the correct number of chromosomes and content of DNA per cell in list-II.

List-I	List-II
A. Prophase of meiosis I	(I) 4n chromosomes and 4C DNA
B. Stage between meiosis I and meiosis II	(II) 2n chromosomes and 4C DNA
C. After cytokinesis of meiosis II	(III) 4n chromosomes and 8C DNA
D. Start of S-phase	(IV) 2n chromosomes and 2C DNA

Choose the most appropriate from the options given below.
(1) A-IV, B-I, C-II, D-III
(2) A-III, B-II, C-IV, D-I
(3) A-III, B-I, C-II, D-IV
(4) A-I, B-IV, C-III, D-II

123. Match the following Columns.

Column I	Column II
A) Calcitonin	1. Treatment of diabetes mellitus
B) Gonadotropin	2. Treatment of Osteoporosis
C) Erythropoietin	3. Diabetes insipidus
D) Insulin	4. Formation of erythrocytes
E) ADH	5.Treatment of infertility

(1) A-3, B-1, C-4, D-2, E-5
(2) A-3, B-2, C-1, D-5, E-4
(3) A-4, B-3, C-2, D-1, E-5
(4) A-2, B-5, C-4, D-1, E-3

124. The two chromatids of a metaphase chromosome represent:
(1) replicated chromosomes to be separated at anaphase
(2) homologous chromosomes of a diploid set
(3) non-homologous chromosomes joined at the centromere
(4) maternal and paternal chromosomes joined at the centromere

125. Correct sequence of stages in the evolution of the modern man, (*Homo sapiens*), is:
(1) Neanderthal man→*Australopithecus*→ Cro-Magnon man→ *Homo erectus*→ Modern man
(2) *Australopithecus*→*Homo erectus*→ Neanderthal man→ Cro-Magnon man→ Modern man
(3) *Homo erectus*→ *Australopithecus*→ Neanderthal man→ Cro-Magnon man→Modern man
(4) *Australopithecus*→ Neanderthal man→ Cro-Magnon man→ *Homo erectus*→ Modern man

126. Photorespiration is favoured by :
(1) low light and high O_2
(2) high O_2 and low CO_2
(3) low temperature and high O_2
(4) low O_2 and high CO_2

127. Match the following w.r.t. frogs

Column-I	Column-II
A. Cranial nerves	I. 10 to 12 in number
B. Vasa efferentia	II. 10 pairs
C. Copulatory pad	III. Organs of touch
D. Sensory papillae	IV. First digit of forelimb

(1) A-I; B-IV; C-II; D-III
(2) A-II; B-I; C-IV; D-III
(3) A-IV; B-I; C-II; D-III
(4) A-IV; B-II; C-I; D-III

128. Which of the following statement is correct ?
(1) C_3 plants respond to higher temperature, show higher photosynthetic rate, while C_4 plants have lower optimum temperature.
(2) Tropical plants have higher temperature optimum than the plants adapted to temperate climate.
(3) Light reaction is not less temperature sensitive than dark reaction.
(4) The effect of water as a factor is less through its effect on plant, rather than directly on photosynthesis.

129. Consider the following two statements.
Assertion(A): Not all copulations lead to fertilisation and pregnancy.
Reason(R): Fertilisation can occur if the ovum and the sperm are simultaneously transported to the ampullary region.
(1) Both A and R are correct, and R correctly explains A.
(2) Both A and R are correct, but R does not explain A.
(3) A is correct but R is incorrect.
(4) R is incorrect, but A is correct.

130. Which of the following is/are correct about amphibians?

I	Skin is moist without scales
II	A tympanum represents the ear
III	Fertilisation is external

(1) Only I and II are correct.
(2) Only I and III are correct.
(3) Only II and III are correct.
(4) I, II, and III are correct

131. Which of the following statement is true?

(1) H_2O, not H_2S, is involved in photosynthesis of purple sulphur bacteria.

(2) Light and dark reactions cannot stop in the absence of light.

(3) Calvin cycle occurs in the grana of chloroplast.

(4) ATP is produced during light reaction via chemiosmosis.

132. Which of the following is not false for oxidative phosphorylation?

(1) It uses oxygen as the initial electron donor.

(2) Energy of oxidation reduction is utilised to synthesize ATP.

(3) It involves an ATP synthase located in thylakoid membrane.

(4) It is independent of chemiosmosis.

133. Consider the following statements about frogs.

Statement I: In water, skin acts as an aquatic respiratory organ (cutaneous respiration).

Statement II: On land, the buccal cavity, skin, and lungs act as the respiratory organs.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

134. Given below are two statements: one is labelled Assertion(A) and the other is labelled as Reason(R):

Assertion (A): Juxtamedullary nephrons are key adaptations for water conservation in mammals.

Reason (R): Juxtamedullary nephrons constitute about 85% of all nephrons in the mammalian kidney.

In the light of the above statements choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false

135. Two redox equivalents are removed (in the form of 2 hydrogen atoms) from ____ and transferred to a molecule of NAD^+ in glycolysis. Fill in the correct blank.

(1) 1,3-biphosphoglyceric acid

(2) 2-phosphoglyceric acid

(3) Phosphoglyceraldehyde-3-phosphate

(4) Phosphoenol pyruvate

136. Consider the following statements.

I: Acetylcholine is released at the electrical synapse.

II: Electrical synapses are faster.

III: Electrical synapses are rare in our system.

Which of the above statements is/are true?

(1) I and II only (2) I and III only

(3) II and III only (4) I, II, and III

137. Select the ***incorrect*** statement among the following.

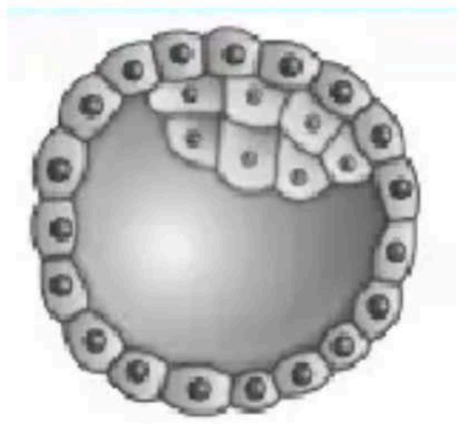
(1) Increase in growth per unit time is growth rate.

(2) A sigmoid growth curve is a characteristic of most living organisms in their natural environment.

(3) Rate of growth is constant during geometrical growth.

(4) Exponential growth can be expressed as $W_1 = W_0e^{rt}$

138. The structure formed during human embryonic development is shown in the figure, given below.



(1) It is morula and has 8 to 16 cells.

(2) It is a tertiary follicle with a fluid-filled antrum.

(3) It is blastocyst that gets implanted in the uterine endometrium.

(4) It is a gastrula, where the three germ layers are formed before the implantation.

139. Consider the following statements about the human skull.

I. It has a total of 22 bones.

II. Cranial bones are 14 in number, and the facial bones are 8 in number.

III. It is monocondylic.

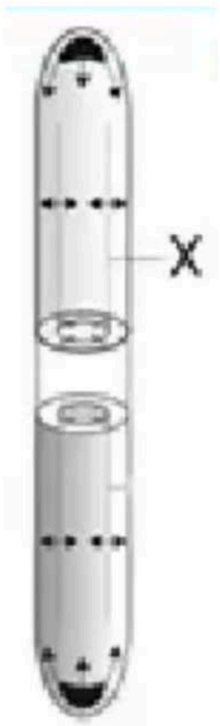
IV. A single hyoid bone is located at the base of the buccal cavity

The correct statements include:

(1) Only I (2) Only I and IV

(3) Only II and III (4) Only I, II, and IV

140. Find the ***incorrect*** statement for X labelled in the diagram:



(1) It helps in increasing the diameter of plant.

(2) It form secondary vascular tissue.

(3) It is meristematic in nature.

(4) It is a kind of tissue present in shoot only.

141. Neural signals through the sympathetic nerves (part of ANS) can increase the

I. rate of heartbeat

II. strength of ventricular contraction

III. cardiac output

(1) Only I and II (2) Only I and III

(3) Only II and III (4) I, II and III

142. Match column I with column II and select the correct combinations from the options given.

Column I (PGRs)	Column II (Derivatives of)
a. Auxin	1. Terpenes
b. Cytokinin	2. Indole compounds
c. Gibberellin	3.Carotenoids derivatives
d. ABA	4.Adenine derivatives

(1) a-2, b-1, c-3, d-4 (2) a-2, b-3, c-4, d-1

(3) a-3, b-2, c-1, d-4 (4) a-2, b-4, c-1, d-3

143. Given below are statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Nephrons are of two types: Cortical and Juxtamedullary, based on their relative position in cortex and medulla of kidney.

Reason (R): Juxta medullary nephrons have a shorter loop of Henle whereas cortical nephrons have a longer loop of Henle.

In the light of the above statements, choose the correct answer from the options given below:

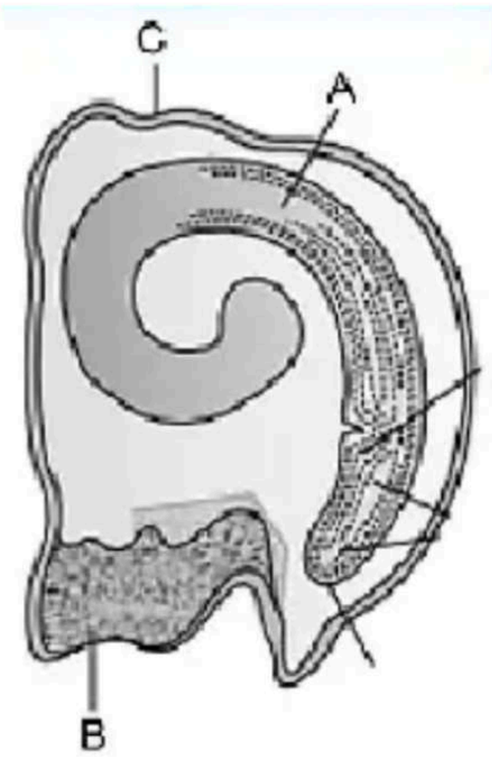
(1) (A) is false but (R) is true.

(2) Both (A) and (R) are true and (R) is the correct explanation of (A).

(3) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(4) (A) is true but (R) is false

144. Select the correct description w.r.t labelling of given diagram:



(1) A = Single cotyledon.

(2) B = Triploid, residual nutritive tissue

(3) C = pericarp

(4) More than one correct

145. Consider the two statements.

Statement I: Any deviation from the normal shape of ECG indicates a possible abnormality or disease.

Statement II: The ECGs obtained from different individuals have roughly the same shape for a given lead configuration.

(1) Both statement I and statement II are incorrect.

(2) Statement I is correct, but statement II is incorrect.

(3) Both statement I and statement II are correct.

(4) Statement I is incorrect, but statement II is correct.

146. Statement I: Frogs have the ability to change their body colour to camouflage themselves from predators.

Statement II: They undergo summer sleep called hibernation as well as winter sleep called aestivation.

(1) Only Statement II is correct.

(2) Only Statement I is correct.

(3) Both Statement I and II are incorrect.

(4) Both Statement I and II are correct.

147. Choose the correctly matched statements from the following ARTs (Assisted Reproductive Technologies).

I. IUT–Embryo is transferred into the uterus after *in vitro* fertilisation.

II. GIFT–Only male gametes are collected and injected into the female body.

III. AI–Taking semen from donor and transferring it into the female genital tract.

IV. ZIFT–Embryo transferred into the fallopian tube.

(1) Only I, II and III

(2) Only I, III and IV

(3) Only III and IV

(4) I, II, III and IV

148. Read the following statements (assertion-A and reason-R) and select the correct option.

A: Apomixis is a form of asexual reproduction that mimics sexual reproduction.

R: Apomixis is a special mechanism to produce fruits without fertilisation.

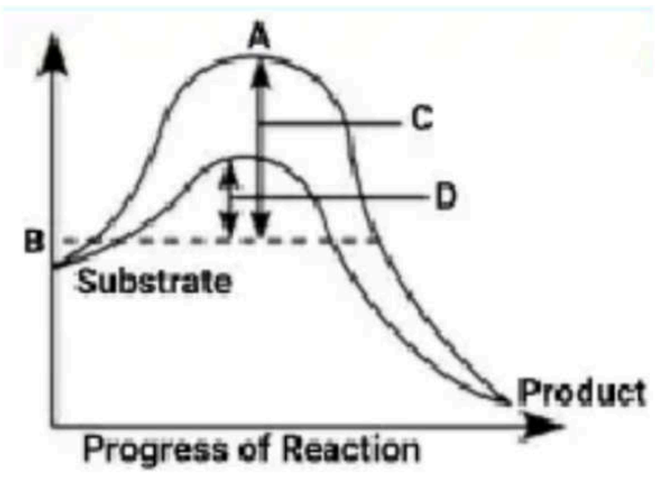
(1) Both Assertion(A) and Reason(R) are true, and Reason(R) is a correct explanation of Assertion(A).

(2) Both Assertion(A) and Reason(R) are true, and Reason(R) is a correct explanation of Assertion(A).

(3) Assertion (A) is true, but Reason (R) is false.

(4) Both A and R are false statements

149. The figure given below shows the conversions of a substrate into product by an enzyme. In which one of the four options (1-4), the components of the reaction labelled as A, B, C and D are identified correctly:



	A	B	C	D
(1)	Transition state	Potential energy	Activation energy without enzyme	Activation energy with enzyme
(2)	Potential energy	Transition state	Activation energy with enzyme	Activation energy without enzyme
(3)	Activation energy with enzyme	Transition state	Activation energy without enzyme	Potential energy
(4)	Potential energy	Transition state	Activation energy without enzyme	Activation energy with enzyme

150. Statement-I: Enzyme substrate complex is short-lived.
Statement-II: Cofactors are bound to the enzyme to make the enzyme catalytically active.
(1) Both Statement I and II are correct.
(2) Both Statement I and II are incorrect.
(3) Statement I is correct and II is incorrect.
(4) Statement I is incorrect and II is correct.

151. Statement 1: Insulin for curing diabetes mellitus should be extracted only from animals.
Statement 2: Insulin has three peptide chains in its active stage.
(1) Statement 1 is correct
(2) Statement 2 is correct
(3) Both are correct
(4) Both are incorrect

152. Assertion(A): Cellulose and chitin both are homopolymers.
Reason (R): Cellulose is made up of identical monomers while chitin is made up of variable monomer units.
(1) Both Assertion(A) and Reason(R) are true, and Reason(R) is a correct explanation of Assertion(A).
(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
(3) Assertion (A) is true, but Reason (R) is false.
(4) Assertion (A) is false, but Reason (R) is true.

153. Study the given statements and select the correct option.
(i) Glycerol is trihydroxy propane.
(ii) Insulin is a homopolymer of fructose.
(iii) Starch gives blue colour with iodine.
(iv) Polysaccharides are long chains of sugars.
(1) Statements (i), (ii) and (iii) are correct.
(2) Statements (i), (iii) and (iv) are correct.
(3) Statements (ii) and (iii) are correct.
(4) All statements are correct.

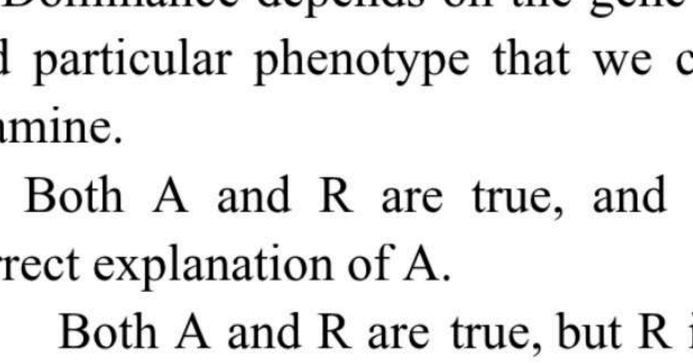
154. How many of the following statements are ***incorrect***?
I. In the organic farming, bio control methods are based on the ability of predator to regulate prey population.
II. A key belief of the organic farmer is that biodiversity furthers health.
III. The conventional farming practices rarely used chemical methods.
IV. The organic farmer holds the view that the eradication of the pests is not only possible, but also desirable.
(1) 4 (2) 3
(3) 2 (4) 1

155. Statement I: The neural system is composed of highly specialised cells called nephrons that can detect stimuli.
Statement II: The neural system of insects includes the brain along with a number of ganglia and neural tissues.
(1) Statement I and II both are correct.
(2) Statement I is correct, but II is incorrect.
(3) Statement I is incorrect, but II is correct.
(4) Statement I and II are both incorrect

156. Given below are two statements.
Statement I : Independent assortment is shown by the genes present on different chromosomes
Statement II : Linkage is shown by the genes present on same chromosome.
(1) Both statement I and II are incorrect.
(2) Statement I is correct but II is incorrect.
(3) Statement I is incorrect but II is correct.
(4) Both statement I and II are correct.

157. Read the following statements (assertion-A and reason-R) and select the correct option.
A: Dominance is not an autonomous feature of a gene.
R: Dominance depends on the gene products and particular phenotype that we choose to examine.
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true but R is false.
(4) Both A and R are false statements

158. The accompanying pedigree shows two individuals who are affected with Phenyl ketoneuria (PKU). What is the probability to be their grand child affected with this disorder :



(1) 25 %
(2) 50 %
(3) 12.5 %
(4) 0 %

159. Which of the following type of gamete will not be produced by a genotype AAbbCcDd ?
(1) AbCd
(2) AbCD
(3) AbcD
(4) AbCcd

160. Statement I: The template/antisense strand of DNA transcribes into RNA.
Statement II: Promotor is situated at 5' end (upstream) of the structural gene w.r.t. polarity of coding strand.
(1) Both statements are correct.
(2) Statement I is correct and II is incorrect.
(3) Statement I is incorrect and II is correct.
(4) Both statements are incorrect.

161. Competition may
(a) lead to organic evolution.
(b) lead to exclusion but not co-existence of species.
(c) occur between closely related and unrelated species.
(d) lowers the intrinsic rate of increase (r) of one species.
The correct one is/are
(1) (d) only
(2) (a), (c), (d) only
(3) (a), (b), (d) only
(4) (c) only

162. What is the correct sequential order of discoveries/events among given below?
a. First identification of DNA
b. Proposition of double helix model of DNA
c. Transformation principle
d. Unequivocal proof that DNA is genetic material
e. Starting of HGP
(1) a → b → c → d → e
(2) b → d → e → a → c
(3) a → c → d → b → e
(4) a → d → c → e → b

163. Which statement is ***incorrect*** for VNTRs ?
(1) Copy number of VNTR varies from chromosome to chromosome in an individual.
(2) VNTRs are not translated.
(3) Size of VNTRs varies from 0.1-20 kb.
(4) VNTR belongs to a class of satellite DNA referred to as microsatellite.

164. How many nucleotides should be present in the translation unit of a processed mRNA if it codes for a polypeptide chain containing 120 amino acids?
(1) 120 (2) 121
(3) 363 (4) 360

165. Match the following columns and choose the correct option.

Column I	Column II
a. Predation	(i) Clown fish-Sea anemone
b. Competition	(ii) Sparrow-seed
c. Commensalism	(iii) Yucca-Moth
d. Mutualism	(iv) *Balanus-Chathamalus*

(1) a(ii), b(iv), c(i), d(iii)
(2) a(ii), b(i), c(iv), d(iii)
(3) a(iii), b(ii), c(iv), d(i)
(4) a(iii), b(iv), c(i), d(ii)

166. From the following, identify the correct combination of salient features of genetic code :
(1) Non-punctuated, Degenerate, Overlapping
(2) Universal, ambiguous, Overlapping
(3) Universal, Ambiguous, Degenerate
(4) Degenerate, Non-overlapping, Unambiguous

167. Select the ***incorrect*** statement.
(1) Gene regulation is a metabolic, physiological and environmental condition that regulates the expression of gene.
(2) Regulatory proteins only act negatively.
(3) DNA fingerprinting involves identifying differences in specific regions in DNA sequence.
(4) DNA fingerprinting is the basis of paternity testing.

168. Select the incorrect match.

Column I	Column II
(1) Lungs of the planet:	Amazon rain forest
(2) Ex-situ conservation:	Wildlife sanctuaries
(3) Biodiversity hotspot :	Indo-Burma
(4) *Lantana:*	Exotic species

169. Which of the following statements is *incorrect*?
(1) UTRs are required for efficient translation process
(2) The split gene arrangement represents the advanced feature of the genome.
(3) The process of splicing represents the dominance of RNA-world.
(4) The presence of introns in split gene is reminiscent of antiquity.

170. In most ecosystems, the pyramids of number, energy and biomass, all are upright due to all the following except
(1) energy at a lower trophic level is always more than at a higher level.
(2) producers are more in number and biomass than herbivores.
(3) biomass of carnivores is more than herbivores
(4) herbivores are more in number than carnivores.

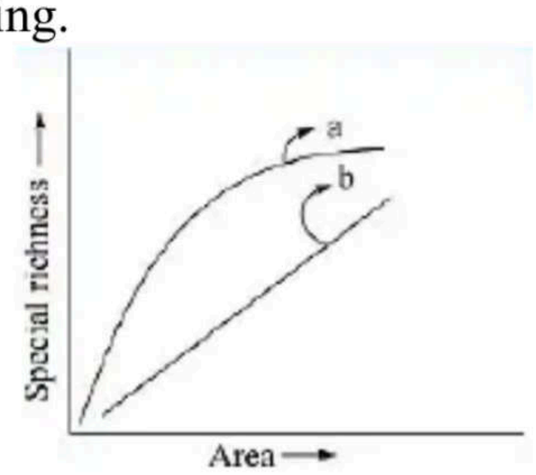
171. Which of the following is *incorrect*?
(1) Amount of all the inorganic substances present in an ecosystem per unit area at a given time is called standing crop.
(2) Fragmentation, leaching and catabolism are the steps of decomposition that occur simultaneously on detritus.
(3) Ecosystem follows both first and second law of thermodynamics
(4) Small amount of energy (2-10% PAR) sustains the entire living world

172. Stirred-tank bioreactors have been designed for:
(1) availability of oxygen throughout the process
(2) ensuring anaerobic conditions in the culture
(3) purification of product
(4) addition of preservatives to the product

173. Which of the following is not a component of downstream processing?
(1) Preservation (2) Expression
(3) Separation (4) Purification

174. Which of the following is referred to as the "Evil Quartet" with reference to loss of biodiversity?
(1) Species richness, extinctions, deforestation, erosion
(2) Habitat loss and fragmentation, over-exploitation, alien species invasion, co-extinction
(3) Overexploitation, grazing, decomposition, extinction
(4) Habitat destruction, co-extinction, deforestation, species richness

175. Identify the figure and select the correct matching.



(1) a-S = CA^Z ; b - Log C = log S + Z log A
(2) a-Log S = log C + Z log A; S = CAZ
(3) a-S = CA^Z ; b - Log S = log C + Z log A
(4) a-C = SA^Z ; b - Log S = log C + Z log A

176. How many of the following are related to Down syndrome?
Flat back of head, Broad flat face, many "loops" on fingertips, Palm crease, Big and wrinkled tongue, congenital heart disease, gynaecomastia, rudimentary ovaries
(1) 8 (2) 7
(3) 6 (4) 5

177. Which of the following is ***not*** a feature of the plasmids?
(1) Independent replication
(2) Circular structure
(3) Transferable
(4) Single-stranded

178. During genetic manipulation, the desired gene is inserted into the vector by cutting the vector at specific sites for gene insertion. The enzyme used for this purpose is
(1) restriction endonuclease
(2) DNA polymerase
(3) nuclease
(4) ligase

179. Bt cotton is a genetically modified cotton variety containing genes from the bacteria called Bacillus thuringiensis (Bt). Which of the following statements is ***incorrect*** regarding Bt cotton?
(1) It is resistant to the attack of crop pests.
(2) It helps in decreasing the use of pesticides on the field.
(3) It controls cotton boll worms, by the presence of *cryIAc* and *cryIIAb*.
(4) It can increase the population of beneficial insects on the field.

180. Agarose gel electrophoresis is used for:
(1) Construction of recombinant DNA by joining with cloning vectors.
(2) Isolation of DNA molecule.
(3) Cutting of DNA into fragments.
(4) Separation of DNA fragments according to their size